CM 5 problem of discrepancy: (X, R)

X: n elements; R: m sets.

Thm A: $\exists$ a coloring of (X, R) s.t each set of R has discrepancy $O(\sqrt{n \ln m})$.

- take a random coloring
- use multiplicative weight update method.

↓

Sum Discrepancy Algorithm $\left(X = \{v_1, \ldots, v_n\}, \mathcal{R}\right)$

For $t = 1, \ldots, n$:

1. Assign v_t a color that *minimizes* the increase in the total weight of all the sets.
   That is, give v_t a color that realizes the following minimum:

$$\min\left\{\sum_{S\in\mathcal{R}}\left(\text{disc}^{t+1}\left(S \mid v_t \text{ is given color '+1'}\right)\right)^c,\ \sum_{S\in\mathcal{R}}\left(\text{disc}^{t+1}\left(S \mid v_t \text{ is given color '-1'}\right)\right)^c\right\}.$$

at each step, minimize $\sum_S [\text{disc}^t(S)]^c$.

for $c \simeq \ln m$, it gave Thm A.

amazing fact: - $c=1$ does not work [TD 4]

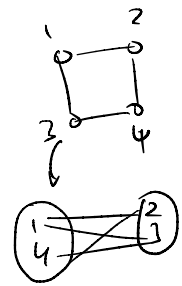
- $c=$ very high does not work [CM 4].

Today: In CM4 → we used a greedy algorithm guided by the function

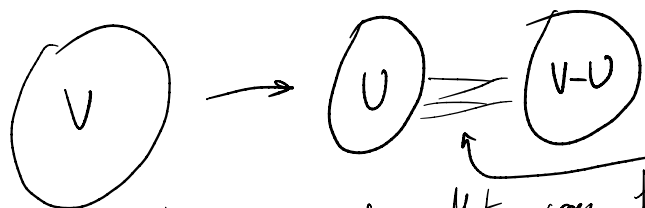
$$\sum_{S \subseteq R} \left[disc^t(S) \right]^c, \quad \text{where } c \approx \ln m.$$

Debur: Given a graph $G = (V, E)$ (n vertices, m edges)

Goal: partition vertices V into 2 sets



eg.

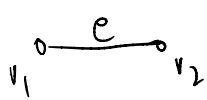


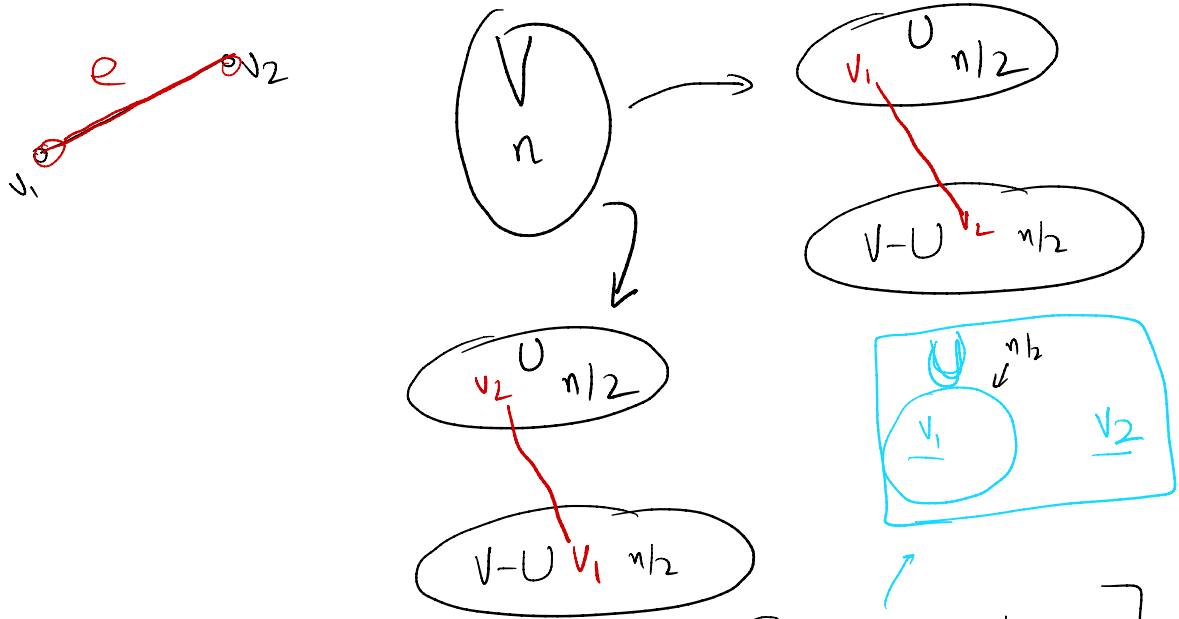
to maximize the # of edges that cross the cut.

Idea: take a random partition over all partitions of V into $\frac{n}{2} - \frac{n}{2}$ sets. → U of size $n/2$ randomly

For an edge $e \in E$. $e = \{v_1, v_2\}$

$$Pr\left[\begin{array}{l} e \text{ crosses this} \\ \text{random cut} \end{array}\right] = Pr\left[\begin{array}{l} v_1 \in U \text{ and} \\ v_2 \notin U \end{array}\right] + Pr\left[\begin{array}{l} v_2 \in U \text{ and} \\ v_1 \notin U \end{array}\right]$$





$$\Pr\left[\begin{array}{l} v_1 \in U \text{ and} \\ v_2 \notin U \end{array}\right] = \frac{\left[\begin{array}{l} \#\text{ of } U \text{ s.t.} \\ v_1 \in U \text{ and } v_2 \notin U \end{array}\right]}{\left[\begin{array}{l} \#\text{ of all} \\ \text{possible } U \end{array}\right]}$$

$$= \frac{\binom{n-2}{\frac{n}{2}-1}}{\binom{n}{\frac{n}{2}}}$$

$$\Pr\left[\begin{array}{c} e \text{ crosses a} \\ \text{random cut} \end{array}\right] = 2 \cdot \frac{\binom{n-2}{\frac{n}{2}-1}}{\binom{n}{n/2}}$$

$$= 2 \cdot \frac{(n-2)!}{\left(\frac{n}{2}-1\right)!\left(\frac{n}{2}-1\right)!} \cdot \frac{\left(\frac{n}{2}\right)!\left(\frac{n}{2}\right)!}{n!}$$

$$= 2 \cdot \frac{\frac{n}{2} \cdot \frac{n}{2}}{n(n-1)} = \frac{1}{2} \cdot \frac{n}{n-1}$$

$$E\left[\begin{array}{c} \#\text{ of edges of } E \\ \text{that cross the cut} \end{array}\right] = \underbrace{m \cdot \frac{1}{2} \cdot \frac{n}{n-1}}$$

slightly more than $\frac{m}{2}$.

Proof 2: deep idea:

instead of picking a random U of size exactly $\frac{n}{2}$, & pick U randomly in a different way:

→ add each vertex of V to U independently with prob. $\frac{1}{2}$.

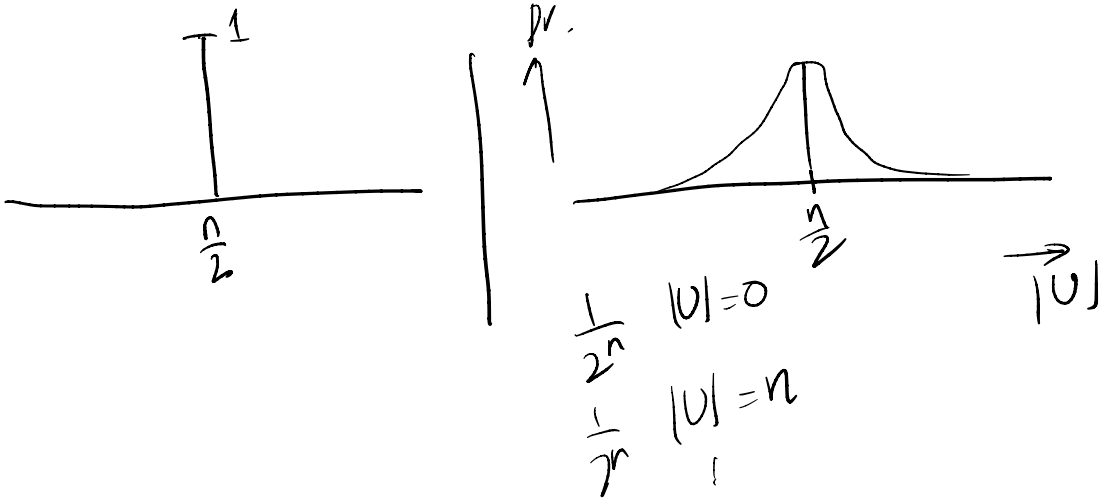
To be clear: these are two diff random distributions.

Pf 1

- pick U uniformly at random from all subsets of V of size $\frac{n}{2}$

Pf 2

- pick U by going over each vertex of V and add to U independently with pro. $\frac{1}{2}$.



U : pick each $v \in V$ ind. with pr. $\frac{1}{2}$.

$$\Pr\left[\begin{array}{c} e=\{v_1,v_2\} \text{ crosses} \\ \text{this random cut} \end{array}\right] = \underbrace{\frac{1}{2}\cdot\frac{1}{2}}_{\substack{v_1\in U \\ v_2\notin U}} + \underbrace{\frac{1}{2}\cdot\frac{1}{2}}_{\substack{v_1\notin U \\ v_2\in U}}$$

$$= \frac{1}{2}.$$

$$\mathbb{E}\left[\begin{array}{c} \#\text{ of edges} \\ \text{crossing random} \\ \text{cut} \end{array}\right] = \frac{m}{2}.$$

Pf 1

pick U uniformly at random from all subsets of V of size $\frac{n}{2}$

Pf complicated

$$E[\ ] = \frac{n}{2} \cdot \frac{n}{n-1}$$

Pf 2

- pick U by going over each vertex of V and add to U independently with pro. $\frac{1}{2}$.

Pf trivial

$$E[\ ] = \frac{n}{2}.$$

Lesson: rather than fixing the size, averaging over all sizes makes the pf. much simpler!

In CM4 → we used a greedy algorithm guided by the function

$$\sum_{S \in R} \left[\text{disc}^t(S) \right]^c, \quad \text{where } c \approx \ln m.$$

Q: Can we do a similar "trick" here?

YES..

now: follow the greedy algo, but guided by

$$\sum_{S \in R} e^{\eta \cdot \text{disc}^t(S)}$$

here η is a fixed value that we set later.

Taylor exp of e: $e^x = \sum_{i=0}^{\infty} \frac{x^i}{i!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots$

Exp Discrepancy Algorithm 1 $\left(X = \{v_1, \ldots, v_n\}, \mathcal{R}\right)$

For $t = 1, \ldots, n$:

1. Assign v_t a color that realizes the following minimum:

$$\min\left\{\sum_{S\in\mathcal{R}} y^{t+1}\left(S \mid v_t \text{ is given color '+1'}\right), \sum_{S\in\mathcal{R}} y^{t+1}\left(S \mid v_t \text{ is given color '-1'}\right)\right\}.$$

   If both colors give the same total weight, choose either one arbitrarily.

2. Update the weights: for each $S \in \mathcal{R}$, set

$$y^{t+1}(S) = \exp\left(\eta \cdot \operatorname{disc}^{t+1}(S)\right).$$

Pf: $\Omega^t = \sum_{S\in\mathcal{R}} \exp\left(\eta \cdot \operatorname{disc}^t(S)\right)$

v_t: color v_t to minimize Ω^{t+1}.

by averaging, we have

$\exists$ color b v_t

s.t. $\Omega^{t+1} \le \left[\sum_S \exp\left(\eta \operatorname{disc}^{t+1}(S)\right)\right.$ ($v_t \to +1$)

$\left. + \sum_S \exp\left(\eta \operatorname{disc}^{t+1}(S)\right)\right] \cdot \frac{1}{2}$ ($v_t \to -1$)

$$= \frac{1}{2}\left[\sum_{s \in R} \underbrace{\exp(\eta\, disc^{t+1}(s))}_{v_t \to +1} + \underbrace{\exp(\eta\, disc^{t+1}(s))}_{v_t \to -1}\right]$$

$$= \frac{1}{2}\left[\sum_{s \in R} \exp\left(\eta\left(disc^{t}(s)+1\right)\right) + \exp\left(\eta\left(disc^{t}(s)-1\right)\right)\right]$$

$$= \frac{1}{2}\left[\left(e^{\eta}+e^{-\eta}\right)\sum_{s \in R} \exp\left(\eta\, disc^{t}(s)\right)\right]$$

$$= \left(\frac{e^{\eta}+e^{-\eta}}{2}\right)\Omega^{t}$$

$$e^{x} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} \cdots$$

$$\leq 1 + x + x^2$$

$$e^{\eta} \approx 1 + \eta + \eta^2$$

$$e^{-\eta} \approx 1 - \eta + \eta^2$$

$$\leq \frac{2+2\eta^2}{2} \Omega^t = (1+\eta^2)\Omega^t$$

$$\vdots$$

$$\Omega^{n+1} \leq \underbrace{(1+\eta^2)^n}_{1+x\leq e^x} \cdot m$$

$$\exp(\eta\, disc(s)) \leq \sum_s \exp(\eta\, disc(s)) = \Omega^{n+1} \leq e^{\eta^2 n} m$$

taking ln

$$\eta\, disc(s) \leq \ln m + \eta^2 n$$

$$\Rightarrow \quad disc(s) \leq \frac{\ln m}{\eta} + \eta n$$

we set η to minimize the above.

$$\eta = \sqrt{\frac{\ln m}{n}}$$

$$\Rightarrow \quad \mathrm{disc}(S) \leq O\left(\sqrt{n \ln m}\right)$$

$$\Omega^{t+1} \le \frac{1}{2}\left[\sum_s \exp(\eta[\text{disc}^t + 1]) + \exp(\eta(\text{disc}^t(s) - 1))\right]$$

$$= \frac{e^{\eta} + e^{-\eta}}{2} \cdot \Omega^t \le (1+\eta^2)\Omega^t \le e^{\eta^2} \cdot \Omega^t$$

$$\Rightarrow \Omega^{n+1} \le e^{n\cdot\eta^2} \cdot m$$

$$\Rightarrow \exp(\eta \ \text{disc}(s)) \le e^{n\cdot\eta^2} m$$

$$\Rightarrow \text{disc}(s) \le \frac{\ln m}{\eta} + \eta n$$

$$\Rightarrow \text{set } \eta = \sqrt{\frac{\ln m}{n}} \qquad O(\sqrt{n \ln m})$$